

unemployment-analysis

February 2, 2024

1 Unemployment Analysis

This project delves into the intricate landscape of unemployment in India, employing a data-driven approach to unravel insights and patterns. The analysis utilizes two datasets - one providing an overview of unemployment and the other offering a detailed breakdown up to November 2020. The project unfolds in several key tasks, each contributing to a comprehensive understanding of the unemployment scenario in the country.

1: Importing Libraries and Loading Data

In this task, you import necessary libraries such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, and Plotly Express. Then, you load two datasets (df and df2) containing information about unemployment in India.

```
[46]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
import plotly.express as px
import calendar
```

```
[44]: df = pd.read_csv("/content/Unemployment in India.csv")
df2 = pd.read_csv('/content/Unemployment_Rate_upto_11_2020.csv')
```

2: Exploratory Data Analysis (EDA) with Pandas and Seaborn

Here, You will perform initial exploration of the data. You check a sample of 5 rows, identify missing values, check for duplicates, and visualize the distribution of data using pie charts, bar charts, and a pair plot with Seaborn.

```
[4]: df.sample(5)
```

```
[4]:
```

	Region	Date	Frequency	Estimated Unemployment Rate (%)	\
354	West Bengal	29-02-2020	Monthly	3.78	
214	Meghalaya	30-04-2020	Monthly	8.38	
693	Tripura	29-02-2020	Monthly	33.88	
196	Maharashtra	31-12-2019	Monthly	3.03	
25	Assam	30-06-2020	Monthly	0.00	

	Estimated Employed	Estimated Labour Participation Rate (%)	Area
354	26093009	49.51	Rural
214	803118	48.83	Rural
693	415339	66.04	Urban
196	26357625	50.98	Rural
25	7544937	34.38	Rural

```
[5]: df.isnull().sum()
```

```
[5]: Region          0
     Date            0
     Frequency       0
     Estimated Unemployment Rate (%)  0
     Estimated Employed  0
     Estimated Labour Participation Rate (%)  0
     Area            0
     dtype: int64
```

```
[6]: df.duplicated()
```

```
[6]: 0      False
     1      False
     2      False
     3      False
     4      False
     ...
     735    False
     736    False
     737    False
     738    False
     739    False
     Length: 740, dtype: bool
```

```
[7]: df.shape
```

```
[7]: (740, 7)
```

```
[8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 740 entries, 0 to 739
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Region                740 non-null   object
1   Date                  740 non-null   object
2   Frequency              740 non-null   object
```

```

3     Estimated Unemployment Rate (%)      740 non-null    float64
4     Estimated Employed                    740 non-null    int64
5     Estimated Labour Participation Rate (%) 740 non-null    float64
6     Area                                740 non-null    object
dtypes: float64(2), int64(1), object(4)
memory usage: 40.6+ KB

```

```
[9]: df.describe()
```

```

[9]:      Estimated Unemployment Rate (%)  Estimated Employed \
count                                740.000000      7.400000e+02
mean                                 11.787946      7.204460e+06
std                                  10.721298      8.087988e+06
min                                  0.000000      4.942000e+04
25%                                  4.657500      1.190404e+06
50%                                  8.350000      4.744178e+06
75%                                  15.887500      1.127549e+07
max                                  76.740000      4.577751e+07

      Estimated Labour Participation Rate (%)
count                                740.000000
mean                                 42.630122
std                                  8.111094
min                                  13.330000
25%                                  38.062500
50%                                  41.160000
75%                                  45.505000
max                                  72.570000

```

```
[10]: df.columns
```

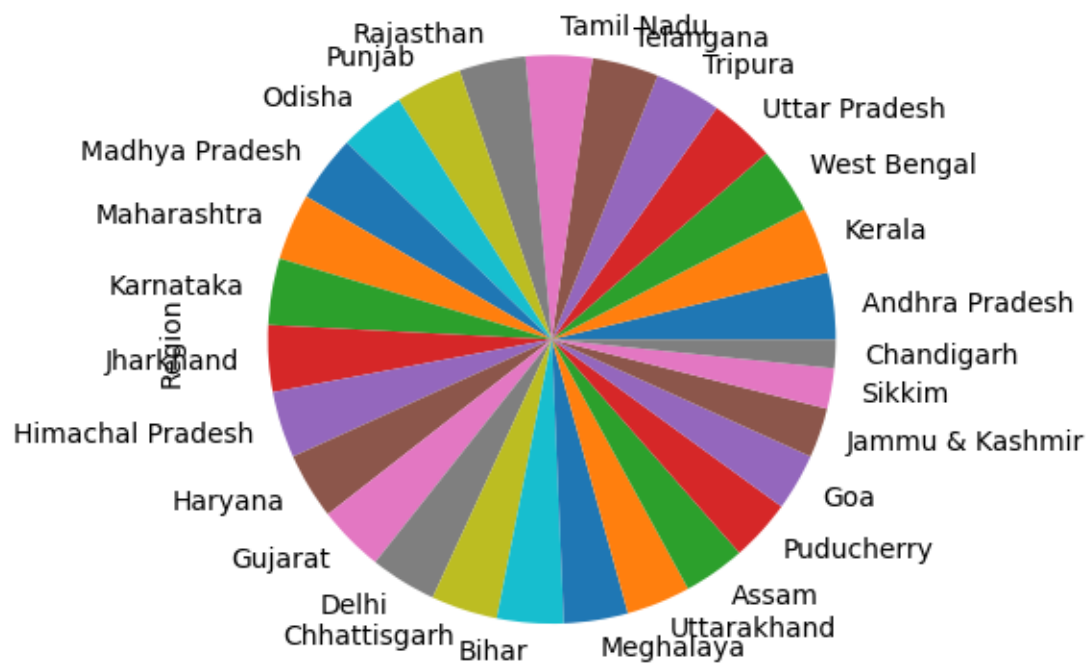
```

[10]: Index(['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate (%)',
          ' Estimated Employed', ' Estimated Labour Participation Rate (%)',
          'Area'],
          dtype='object')

```

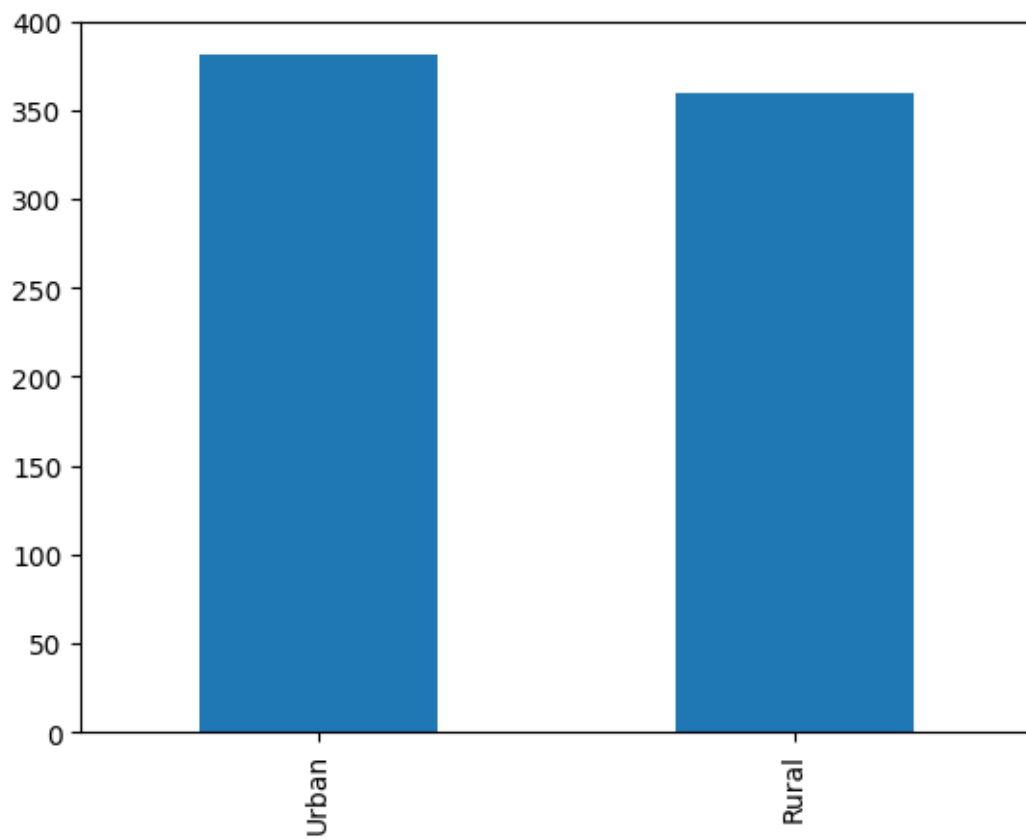
```
[11]: df['Region'].value_counts().plot.pie()
```

```
[11]: <Axes: ylabel='Region'>
```



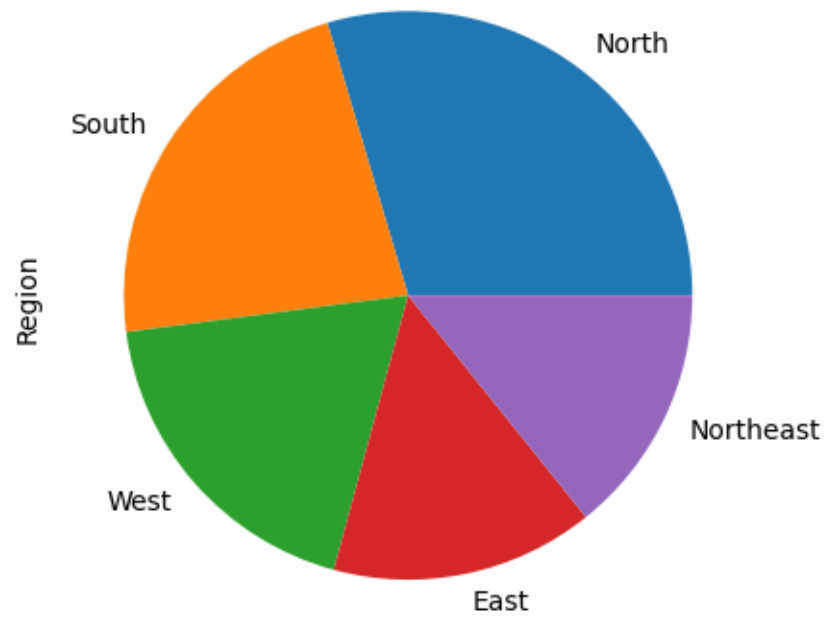
```
[14]: df['Area'].value_counts().plot.bar()
```

```
[14]: <Axes: >
```



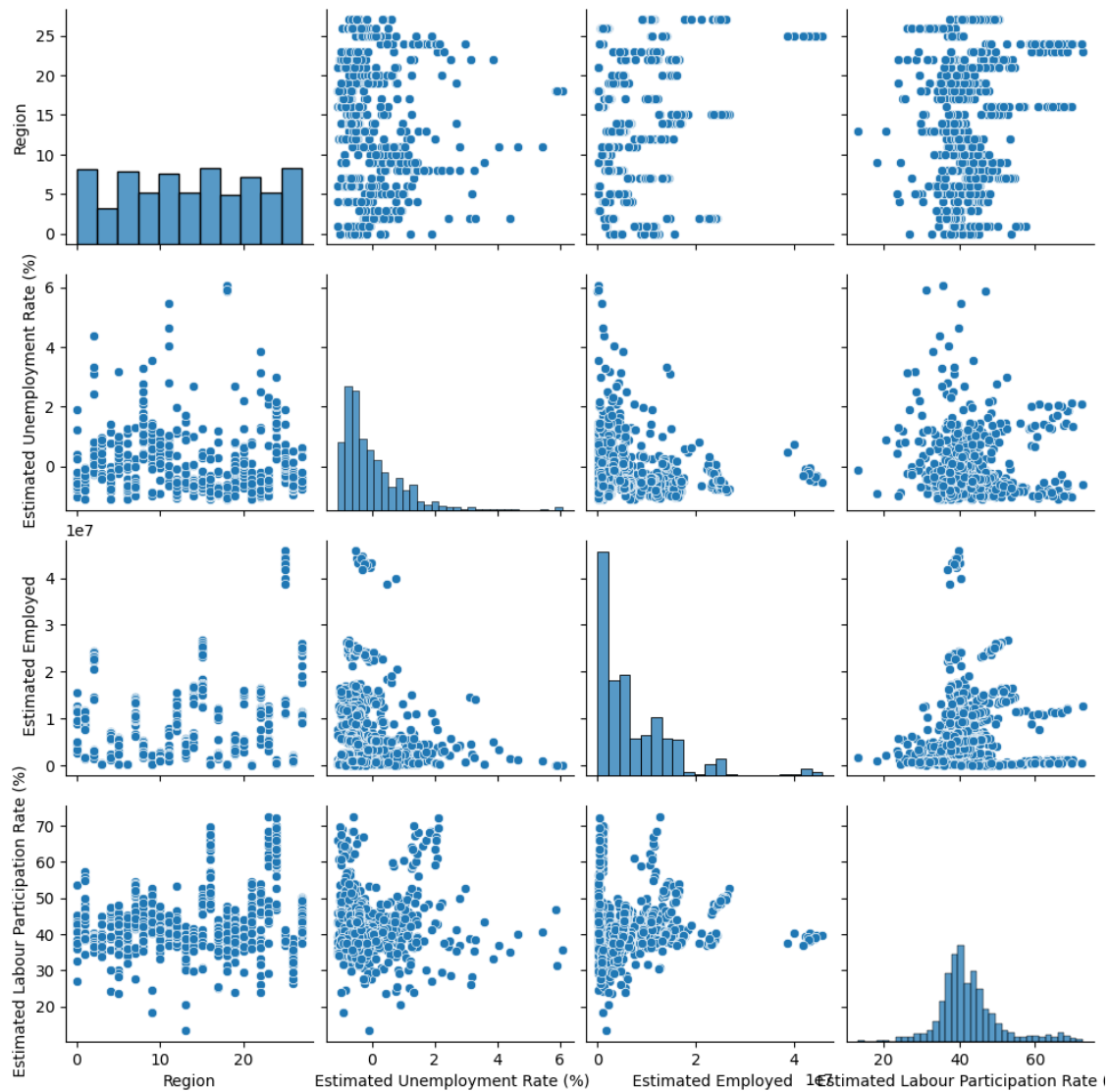
```
[51]: df2['Region'].value_counts().plot.pie()
```

```
[51]: <Axes: ylabel='Region'>
```



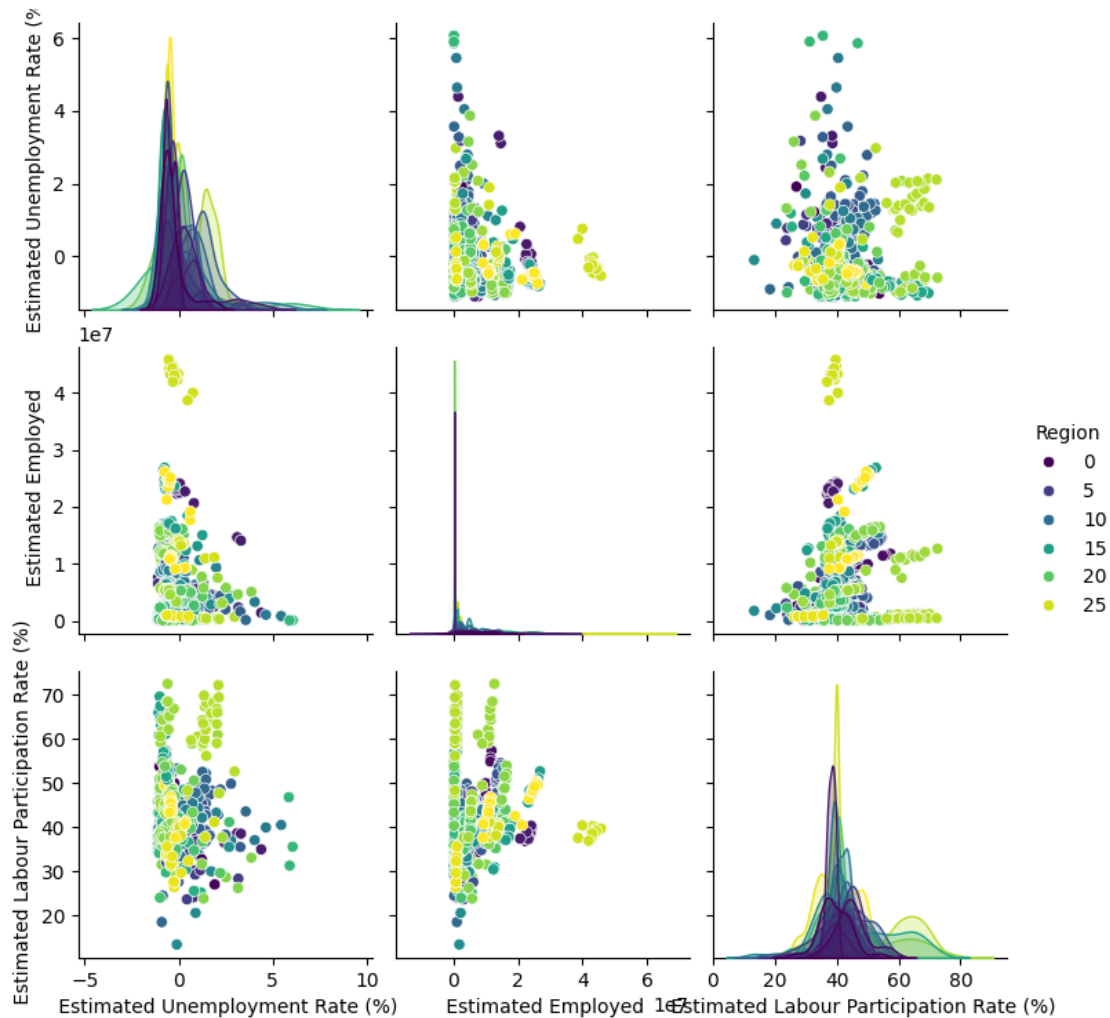
```
[23]: sns.pairplot(df)
```

```
[23]: <seaborn.axisgrid.PairGrid at 0x7fbcf18a6b90>
```



```
[26]: sns.pairplot(df, hue='Region', palette='viridis')
```

```
[26]: <seaborn.axisgrid.PairGrid at 0x7fbcefc8a60>
```



3: Histograms of Estimated Employment and Unemployment Rates

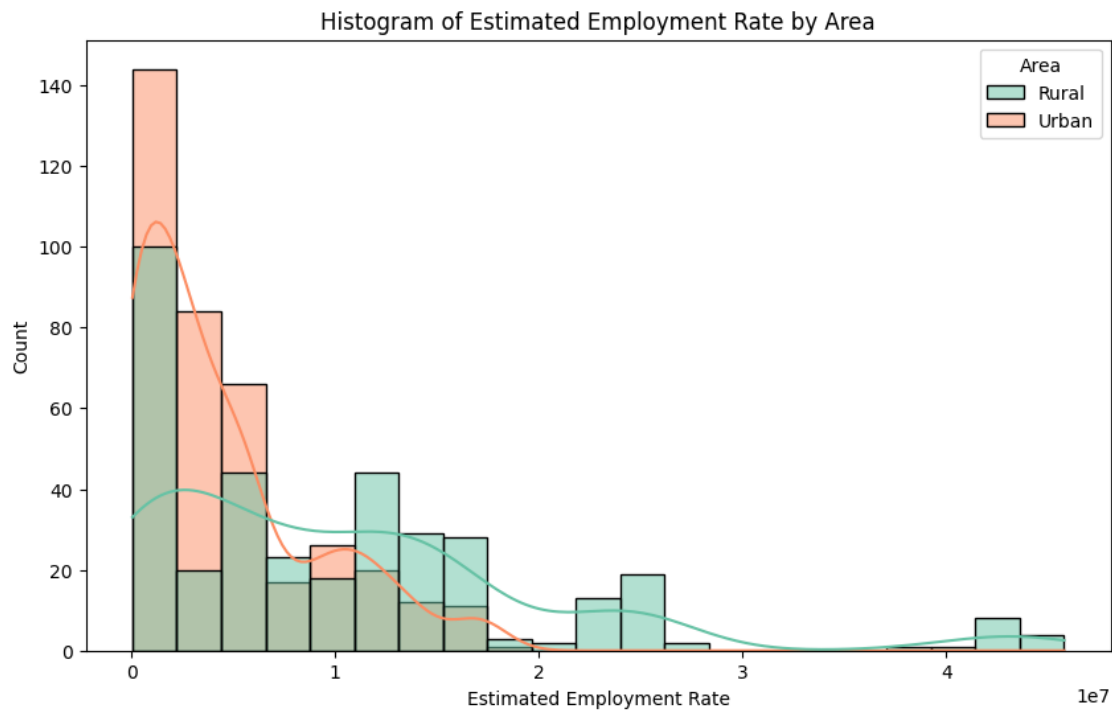
```
[37]: # Histogram of Estimated Employment Rate by Area

plt.style.use("default")

df.columns = ['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate_
↳ (%) ',
              ' Estimated Employed', ' Estimated Labour Participation Rate (%)',
              'Area']
plt.figure(figsize=(10, 6))
sns.histplot(x=' Estimated Employed', hue="Area", data=df, kde=True,
↳ palette="Set2")
plt.title("Histogram of Estimated Employment Rate by Area")
plt.xlabel("Estimated Employment Rate")
plt.ylabel("Count")
```



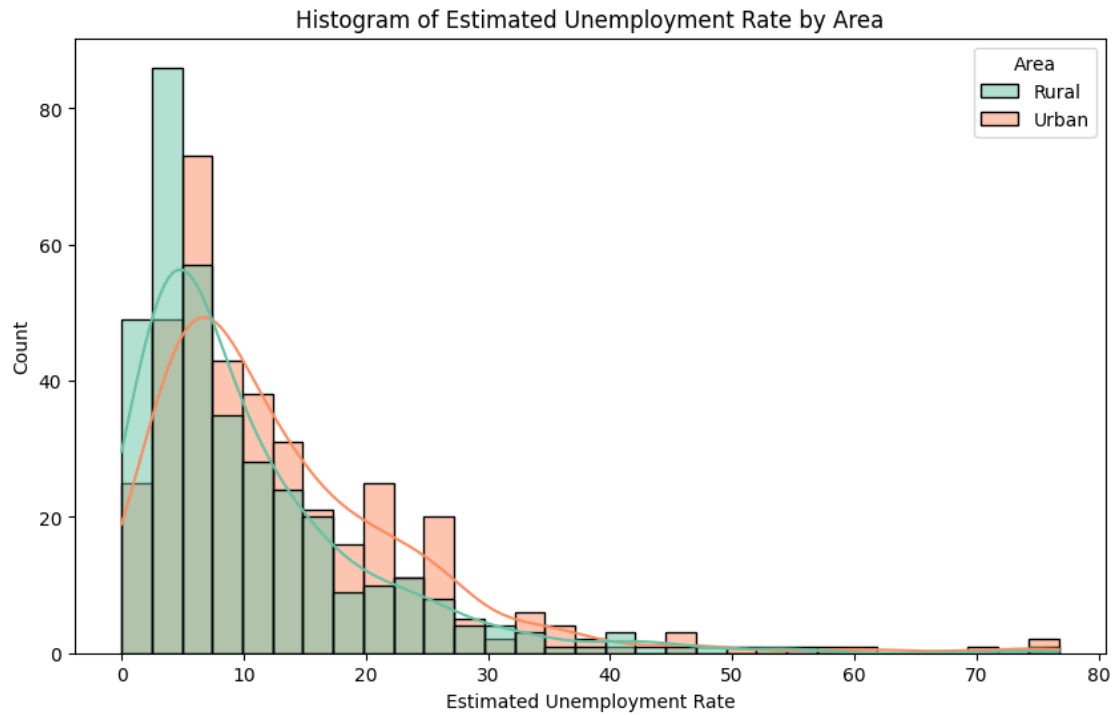
```
plt.show()
```



```
[40]: # Histogram of Estimated Unemployment Rate by Area

plt.style.use("default")

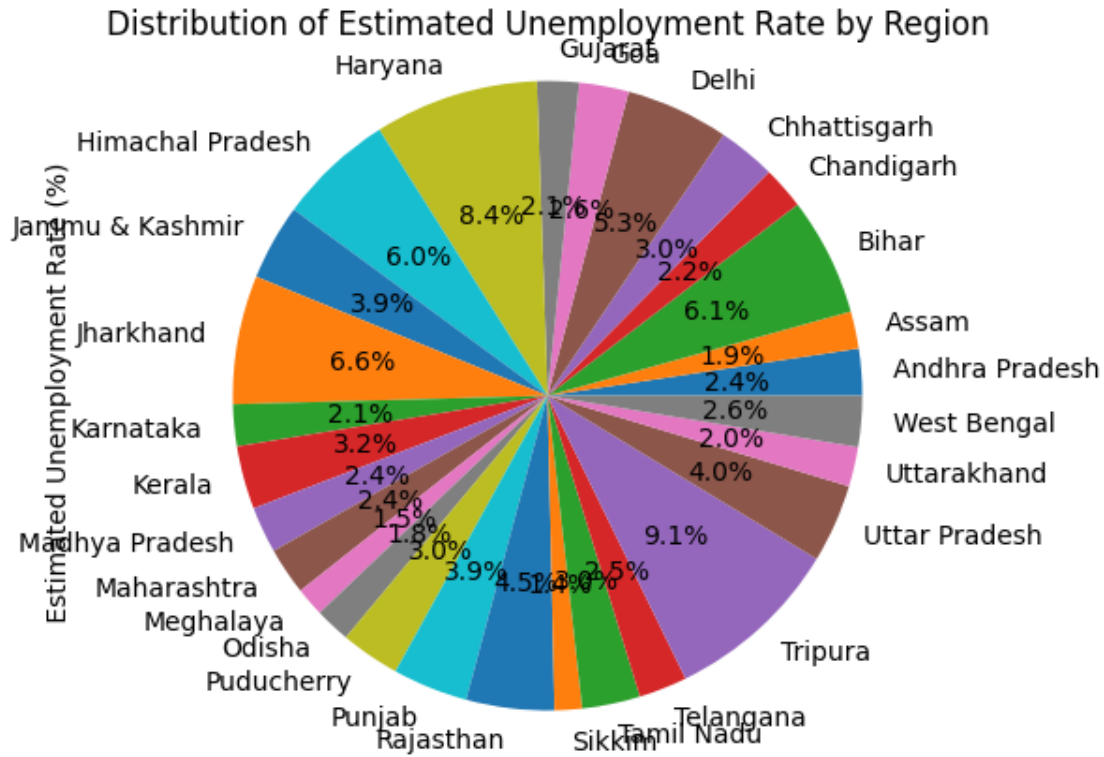
df.columns = ['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate_
↳ (%)',
              ' Estimated Employed', ' Estimated Labour Participation Rate (%)',
              'Area']
plt.figure(figsize=(10, 6))
sns.histplot(x=' Estimated Unemployment Rate (%)', hue="Area", data=df,
↳ kde=True, palette="Set2")
plt.title("Histogram of Estimated Unemployment Rate by Area")
plt.xlabel("Estimated Unemployment Rate")
plt.ylabel("Count")
plt.show()
```



4: Distribution of Estimated Unemployment Rate by Region

```
[39]: # Distribution of Estimated UnEmployment Rate by Region

data = df[['Region', ' Estimated Unemployment Rate (%)']]
data_grouped = df.groupby('Region')[' Estimated Unemployment Rate (%)'].sum()
data_grouped.plot(kind='pie', autopct='%1.1f%%')
plt.title('Distribution of Estimated Unemployment Rate by Region')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.show()
```



5: Data Preprocessing and Feature Engineering

```
[45]: df2.columns
```

```
[45]: Index(['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)',
        'Estimated Employed', 'Estimated Labour Participation Rate (%)',
        'Region.1', 'longitude', 'latitude'],
        dtype='object')
```

```
[47]: df2.columns = ['States', 'Date', 'Frequency', 'Estimated Unemployment Rate',
        ↳ 'Estimated Employed',
        'Estimated Labour Participation Rate', 'Region', 'longitude',
        ↳ 'latitude']

# Converting 'Date' column to datetime format
df2['Date'] = pd.to_datetime(df2['Date'], dayfirst=True)

# Converting 'Frequency' and 'Region' columns to categorical data type
df2['Frequency'] = df2['Frequency'].astype('category')
df2['Region'] = df2['Region'].astype('category')

# Extracting month from 'Date' and creating a 'Month' column
```

```

df2['Month'] = df2['Date'].dt.month

# Converting 'Month' to integer format
df2['Month_int'] = df2['Month'].apply(lambda x: int(x))

# Mapping integer month values to abbreviated month names
df2['Month_name'] = df2['Month_int'].apply(lambda x: calendar.month_abbr[x])
df2.drop(columns= 'Month',inplace = True)

```

6: Sunburst Chart for Unemployment Rate in Each Region and State

```

[48]: # Sunburst chart showing unemployment rate in each region and state

unemplo_df = df2[['States', 'Region', 'Estimated Unemployment Rate', 'Estimated_
↳Employed', 'Estimated Labour Participation Rate']]
unemplo = unemplo_df.groupby(['Region', 'States'])['Estimated Unemployment_
↳Rate'].mean().reset_index()
fig = px.sunburst(unemplo, path=['Region', 'States'], values='Estimated_
↳Unemployment Rate',
                  color_continuous_scale='Plasma', title='Unemployment rate in_
↳each region and state',
                  height=650, template='ggplot2')
fig.show()

```

7: Geo Scatter Plot for Impact of Lockdown on Employment

```

[50]: fig = px.scatter_geo(df2,'longitude', 'latitude', color="Region",
                           hover_name="States", size="Estimated Unemployment Rate",
                           ↳
↳animation_frame="Month_name",scope='asia',template='seaborn',title='Impack_
↳of lockdown on Employment across regions')

fig.layout.updatemenus[0].buttons[0].args[1]["frame"]["duration"] = 3000

fig.update_geos(lataxis_range=[5,35], lonaxis_range=[65,↳
↳100],oceancolor="#3399FF",
               showocean=True)

fig.show()

```

[12]: